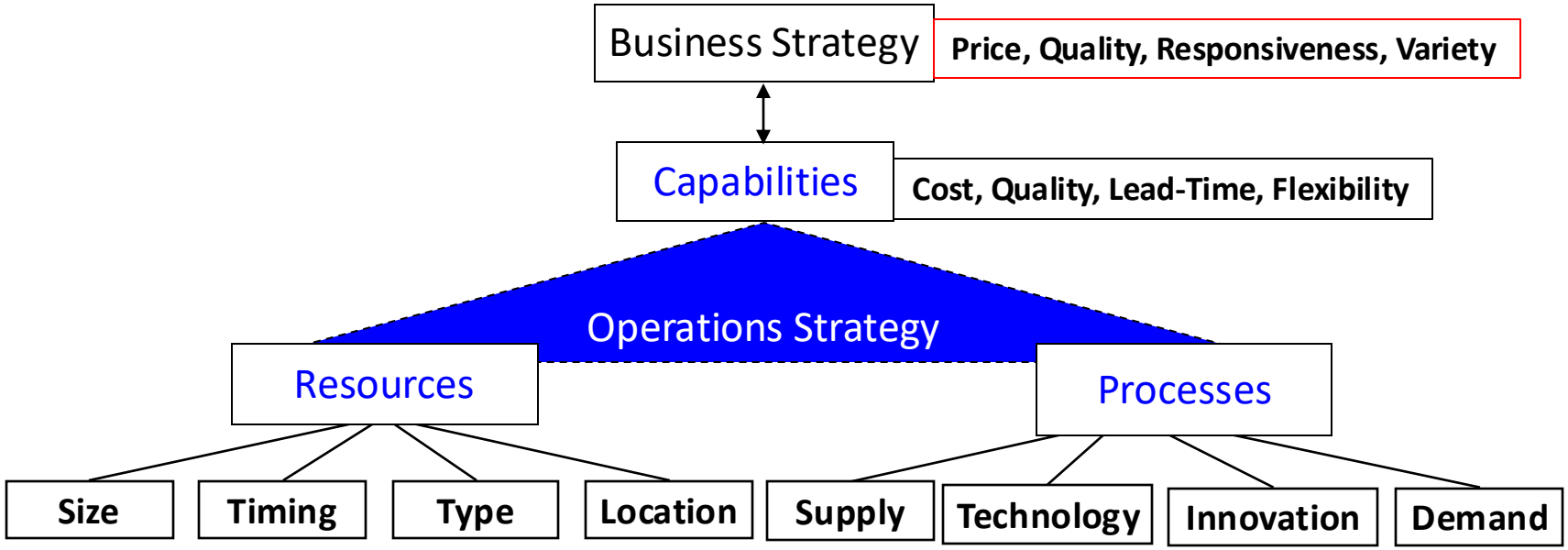
Strategy

- Strategy is a plan (often long term) to reach a particular goal; for business organizations, the goal of strategy is to maximize the Net Present Value (NPV).
- Large organizations do this in a hierarchical manner and specify 3-levels of strategy
  - Corporate strategy is at the highest level, it defines the industries in which the organization will be active, configures business units for each industry, and acquires and allocates corporate resources to each business unit, i.e., a self-contained part of an organization
  - A business unit formulates its business strategy to seek a favorable competitive position in the industry, the fundamental arena in which competition occurs. **Price, quality, responsiveness,** and **variety** are the four different aspects that describes a business strategy
  - To execute its business strategy, an organization further defines functional strategies such as
    - a financial strategy specifying how capital will be raised and invested
    - a marketing strategy specifying how the market will be segmented and how the product will be positioned, and promoted
    - an operations strategy specifying a plan on developing resources and configuring processes for the creation and delivery of products and services (precise definition will be given later)

Operations

- Operations function is responsible for the creation and delivery of products and services. It is helpful to adopt three different views to look at operations -
  1. The process view looks at the transformation process, i.e., activity network, used to create and deliver the products and services
     - It looks at how work or activity is done
       - OS (Operations Strategy) is to make long-term decisions related to the transformation process, e.g., **make or buy, technology adoption and update**, **make to order or stock or how to match supply with demand, etc**.
         - OM (Operations Management) is to manage and improve the day-to-day activities in the transformation process, i.e., it focus on the tactics (course DAO2703)
  2. The resource view looks at the assets used in the transformation process
     - To perform and coordinate activities, organizations need a variety of real assets/resources, resource view looks at who or what do the activities/work, e.g., **size, timing, type, location,** etc.
  3. The competency/capabilities view looks at what operations can and cannot do
     - It begs the important question: which capabilities should an operation have?
       - **Costs, lead-time (LT), flexibility, quality**
       - It should connect to business strategy (on how later)

The Operations Strategy Framework



(Source: Operations Strategy by Prof. Jan A. Van Mieghem)

Competence View of Operations

- Four key aspects to determine capabilities strategically
  1. What **is** the total costs of operating, including fixed and variable costs?
     - especially important in low margin industry
  2. What **is** the ability to deliver quality outputs?
     - Design related such as performance and features
     - Performance related such as reliability and durability
     - Quality is a key differentiator in luxury and high precision businesses
  3. What **is** the total responsiveness or lead-time?
     - key success factor in service and in rapidly changing environment
  4. What **is** operations' flexibility to change?
     - Examples: scope, agility, robustness
     - Note: key success factor in competing on customized products/services

is or should be – connects to business strategy (more later)

(Source: Operations Strategy by Prof. Jan A. Van Mieghem)

Resource View of Operations

- Four key aspects to determine resources strategically
  1. How many resources **should** we invest?
     - Examples: number of plants, capacity size of each plant
     - Note: key trade-off is the cost of excess capacity vs capacity shortfall (opportunity cost);
     - Tool: marginal analysis can be used to determine the optimal capacity size
  2. When **should** we increase or reduce resources?
     - Examples: Eli Lilly abandoned a $325 million insulin plant in 2007; Solar-Panel Maker Solyndra closed a factory and delayed Expansion in 2010, etc.
     - Note: trade-off is the cost of capacity adjustment vs excess/shortfall of capacity
     - Tools: strategic timing and tactical Timing
  3. What kinds/type of resources **are** best?
     - Examples: process type (job shop vs repetitive line), facility and equipment types
     - Note: degree of automation and flexibility is a key decision in operations strategy
  4. Where **should** the resources be located?
     - Examples: 7/11 stores, Zara manufacturing plants
     - Note: globalization is certain to increase

Source: Operations Strategy by Prof. Jan A. Van Mieghem

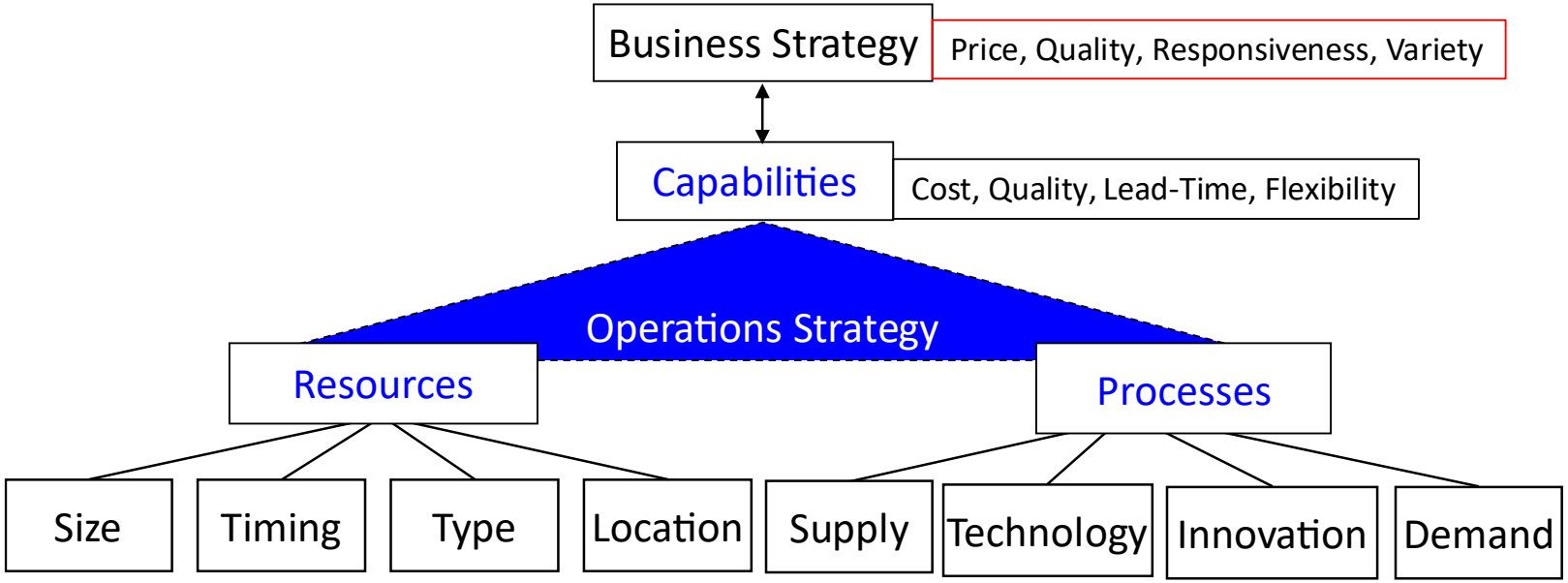
Process View of Operations

Four key aspects to determine processes

1. Which activities should we perform internally, outsourced?
   - This is inputs management, make or buy decision
   - Supply chain examples: vertical integration, outsourcing (multi- or single sourcing)
   - Note: technology affects sourcing decisions and supply chain structure
2. Which technology do our process need?
   - Examples: CAD, CAE, CAM, CAPP, NC, CNC, DNC, CIM, AGVs, Robots, Drones, FMS, GPS, AI, etc.
   - Note: need to link process technology decisions with product technology and product design
   - Case studies: Moderna vs. Eli Lily
3. When and how to improve and innovate?
   - Examples: Continuous improvement (Kaizen, Lean, 6σ quality), radical change (copy best practice, reengineering)
   - Organizational learning is directly related to both improvement and innovation
     - Two activities: conceptual learning (why) and operational learning (how)
     - Two modes: learning before doing (prototyping) and by doing (real production)
4. How to match demand with available supply?
   - This is outputs management, goal is to maximize revenue
   - It is important especially when supply processes cannot quickly adapt to demand changes
   - Tool: price control and capacity control (i.e., Revenue management (RM) which is the collection of demand management practices that aim to maximize the revenue of, especially limited and perishable, available supply)
   - Note: marginal analysis is a useful tool in RM, overbooking is an example of RM

Source: Operations Strategy by Prof. Jan A. Van Mieghem

The concept of Operations Strategy



- Operations Strategy is a long-term plan for developing resources and configuring processes such that the resulting capabilities maximize Net Present Value (NPV)
- Operations Strategy thus answer two main questions:
  1. What should operations be good at, or which capabilities should an organization nurture?
  2. Which operational system of resources and processes best provides these capabilities?

Source: Operations Strategy: principles and practice, by Jan A. Van Mieghem

Alignment

1. What should operations be good at, or which capabilities should it nurture?
   - Answer: connect or align with the business strategy

Business Strategy	What do we compete and provide value to our customers? ➢ Prioritize our customer value proposition around price, responsiveness, quality, variety, etc.
Market perspective	
Competencies	What must operations do particularly well? ➢ Prioritize our process competencies around cost, Lead time, quality, flexibility, etc.
	Resource perspective
Resources & Processes	How do we configure and develop our operations? ➢ Configure the activity network or processes ➢ Develop the bundle of real assets or resources

Source: Operations Strategy by Prof. Jan A. Van Mieghem

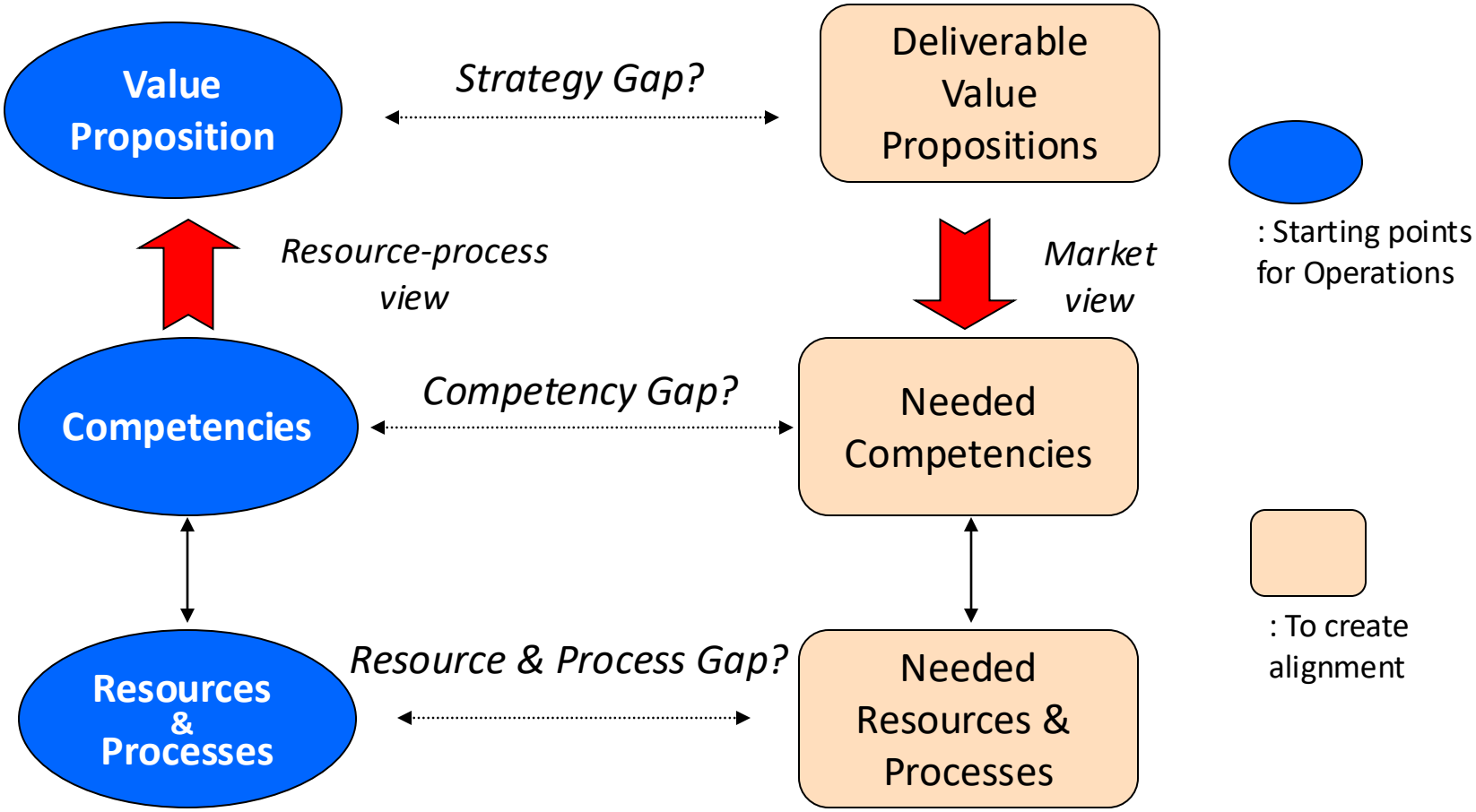
Alignment Principle

- Operations Strategy should develop resources and configure processes such that the resulting capabilities are connected or aligned with the competitive position (defined by business strategy) that a firm seeks over time.
- Introduce two alignment tools
  - Strategic Operational Audit (SOA)
  - Product Process Matrix (PMM)

Source: Operations Strategy by Prof. Jan A. Van Mieghem

Alignment Tool to tailor operations

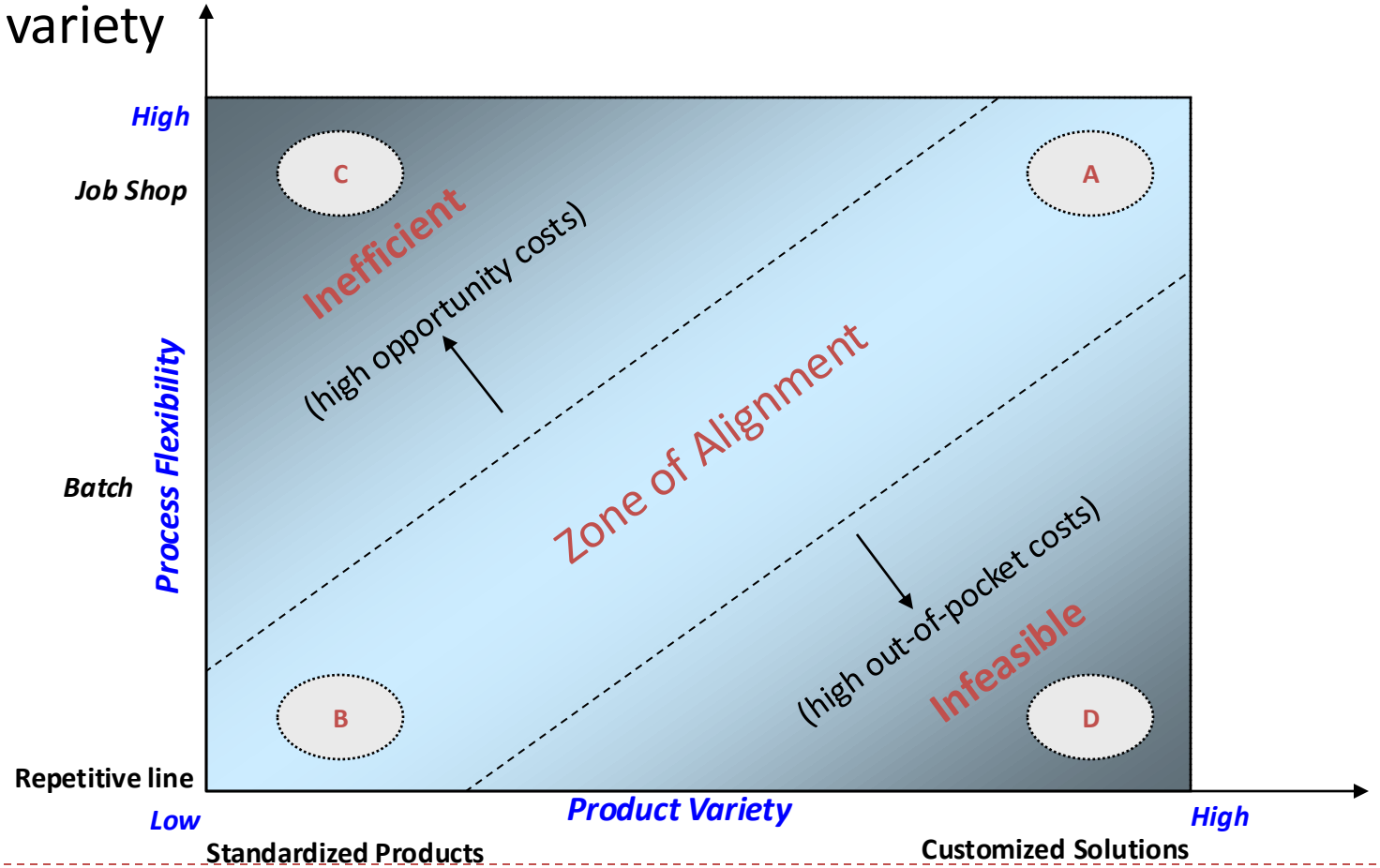
- Strategic Operational Audit (SOA)



(Source: Operations Strategy by Prof. Jan A. Van Mieghem)

Alignment Tool to tailor operations

- Product-Process Matrix (OMM) can be used to match between process type and technology and delivered product attributes and variety



(Source: Operations Strategy by Prof. Jan A. Van Mieghem)

NPV

2. Which operational system of resources and processes best provides these capabilities?
   - The goal is to maximize the NPV (refer to an Excel example on Canvas)
     - NPV = C_o + PV
       - C_o is initial investments costs
       - PV = $\sum_{t=1}^{T} \frac{C_t}{(1+r)^t}$ where
         - r is the rate of return and is called discount rate
         - C_t is return cash flows in future time t

Year (t)	0	1	2	3	4	5	6	7	8	9
Cash return C_t	-$ 400.00	$ 100.00	$ 120.00	$ 110.00	$ 90.00	$ 80.00	$ 100.00	$ 105.00	$ 120.00	$ 85.00
Rate of return r	10%									
Discount 1/(1+r)^t	1	0.91	0.83	0.75	0.68	0.62	0.56	0.51	0.47	0.42
PV	-$ 400.00	$ 90.91	$ 99.17	$ 82.64	$ 61.47	$ 49.67	$ 56.45	$ 53.88	$ 55.98	$ 36.05
NPV	$ 186.23									

Source: Operations Strategy by Prof. Jan A. Van Mieghem

Today's Learning Objectives

- Be able to
  - Adopt three complementary views to describe operations
  - Identify the key decisions and relationships in our framework for operations strategy
  - Explain the concept of operations strategy and its impact on an organization
  - Use tools to tailor operations strategy for alignment
    - SOA & PMM
  - Calculate NPV